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Angles in Polygons and Parallel Lines

Name the angle fact every time you use it. For polygons, start from the exterior angle: they always add to 360° .

THE TOPIC ON ONE MAP

Angles in Polygons and Parallel Lines

01 – Angle facts and parallel lines

alternate, corresponding, co-interior

give a reason for each stage

02 – Interior and exterior angles

n from one angle

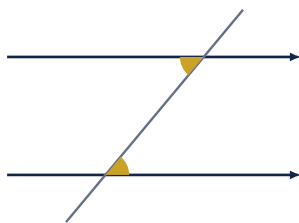
missing angle in a 10-gon

03 – Regular polygons joined together

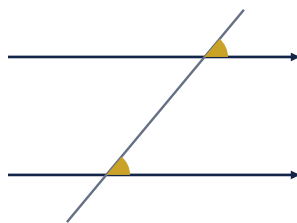
octagon + pentagon

isosceles triangles inside

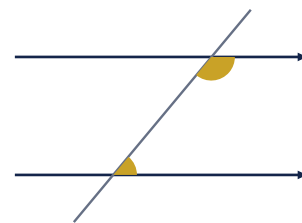
PARALLEL LINES Three pairs to recognise — and the reason you must write:



ALTERNATE (Z)
equal



CORRESPONDING (F)
equal



CO-INTERIOR (C)
add to 180°

ZONE EXTERIOR

sum = 360°

regular: each = $\frac{360}{n}$

ZONE INTERIOR

interior + exterior = 180°

regular: each
= $180 - \frac{360}{n}$

ZONE SUM

$(n - 2) \times 180^\circ$

pentagon 540, hexagon
720

01 Angle Facts and Parallel Lines

Find one angle at a time and write the reason each time. Isosceles triangles hide in the diagram: equal sides mean equal base angles.

WHEN YOU SEE → YOU DO

parallel lines (arrows)

→ alternate / corresponding / co-interior

“give a reason for each stage”

→ write the full angle fact, not just “Z”

equal marks on two sides

→ isosceles: base angles equal

a straight line / a point

→ 180° / 360°

“show that AB is parallel to CD ”

→ find x , show corresponding angles are equal

WORKED MODEL · MAY 2021 P2H

$ABCD \parallel FGHI$; $BE = CE$; angle $BEC = 44^\circ$. Find angle JGH .

1 $EBC = \frac{180-44}{2} = 68^\circ$

base angles of an isosceles triangle

2 $EGH = 68^\circ$

corresponding angles are equal

ANSWER $JGH = 180 - 68 = 112^\circ$ (angles on a straight line)

× **TRAP** Writing only “Z angles”. The mark needs the words: alternate angles are equal.

IDEA 01

Practise: Angle Facts and Parallel Lines

My questions first, then selected practice, then real past papers.

Q1**DR ESLAM**

Parallel lines

CH 4 • TOPIC 2

2 marks

Two parallel lines are cut by a transversal. One angle is 64° . Find the co-interior angle and give the reason.

SELECTED PRACTICE

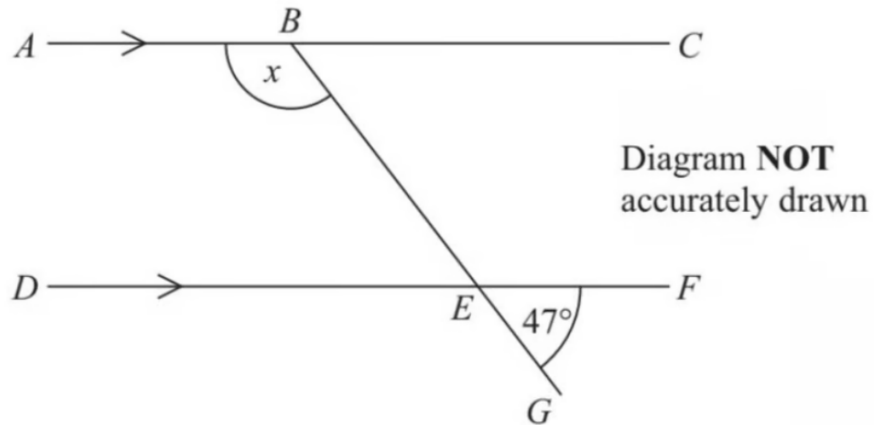
Q2

PRACTICE

Practice · Medium

CH 4 · TOPIC 2 3 marks

1



ABC and DEF are parallel lines.

BEG is a straight line.

Angle $GEF = 47^\circ$.

Work out the size of the angle marked x .

Give reasons for your answer.

(3 marks)

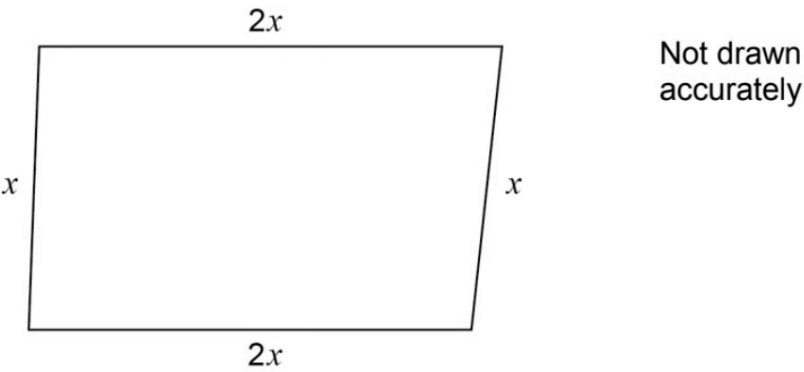
Q3

PRACTICE

Practice • Medium

CH 4 • TOPIC 23 marks

18 Here is a **sketch** of a quadrilateral.
All lengths are in centimetres.



Tick **one** box for each statement.

	True	May be true	Not true
The quadrilateral is a rectangle			
The quadrilateral is a parallelogram			
The quadrilateral is a rhombus			
The quadrilateral is a kite			

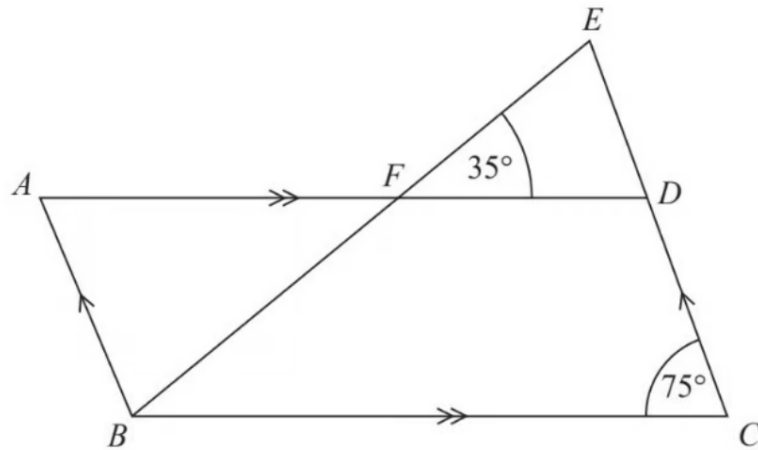
(3 marks)

Q4**PRACTICE**

Practice · Hard

CH 4 · TOPIC 2 4 marks

6



$ABCD$ is a parallelogram.

EDC is a straight line.

F is the point on AD so that BFE is a straight line.

Angle $EFD = 35^\circ$

Angle $DCB = 75^\circ$

Show that angle $ABF = 70^\circ$

Give a reason for each stage of your working.

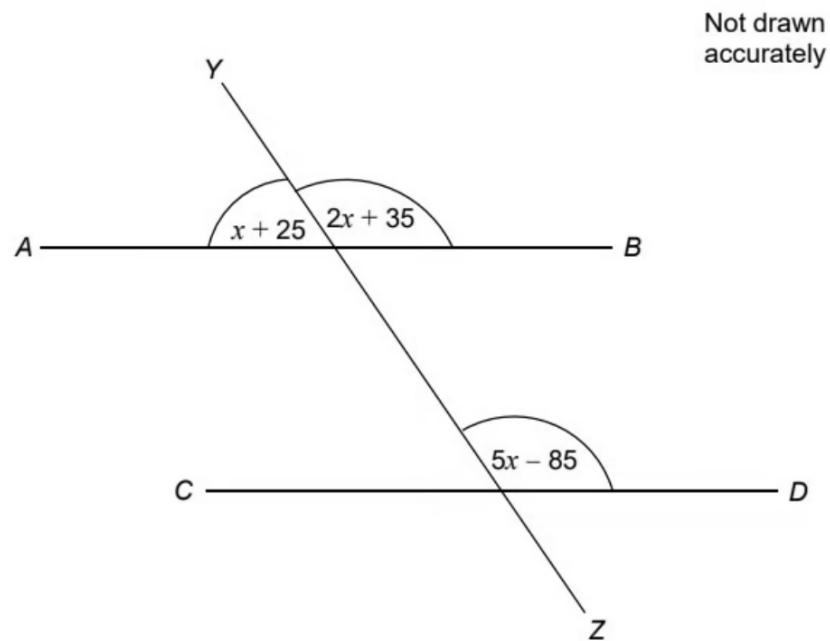
(4 marks)

Q5**PRACTICE**

Practice · Very Hard

CH 4 · TOPIC 2 4 marks

10 AB , CD and YZ are straight lines. All angles are in degrees.



Show that AB is parallel to CD .

(4 marks)

PAST PAPERS

Q6 PAST PAPER Edexcel 4MA1 • January 2020 • Paper 2H • Q4

CH 4 • TOPIC 2 4 marks

- 4 The diagram shows a triangle.

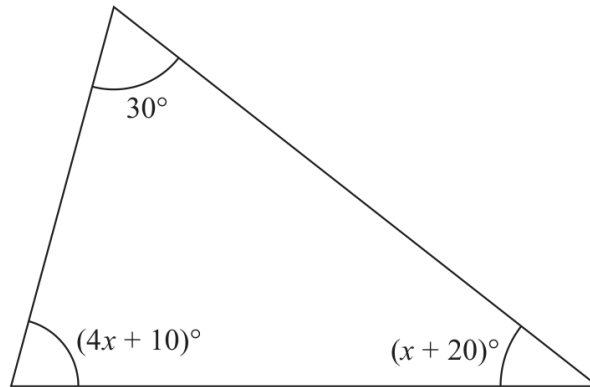


Diagram **NOT**
accurately drawn

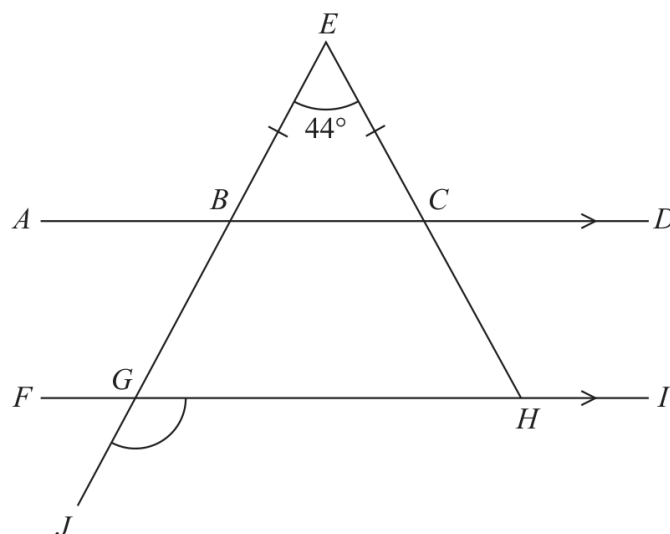
Work out the value of x .

$x =$

(Total for Question 4 is 4 marks)

Q7**PAST PAPER**

Edexcel 4MA1 · May 2021 · Paper 2H · Q5

CH 4 · TOPIC 2 5 marks**5**Diagram **NOT**
accurately drawn*ABCD* and *FGHI* are parallel straight lines.*EBGJ* and *ECH* are straight lines. $BE = CE$ Angle $BEC = 44^\circ$ Work out the size of angle JGH .

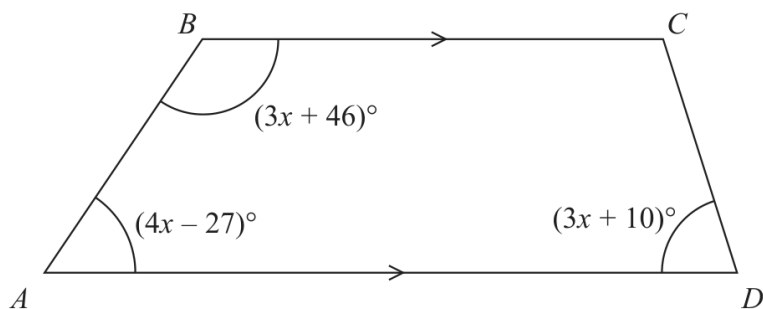
Give a reason for each stage of your working.

○

(Total for Question 5 is 5 marks)

Q8 PAST PAPER Edexcel 4MA1 · June 2023 · Paper 2H · Q3

CH 4 · TOPIC 2 4 marks

3 $ABCD$ is a trapezium.Diagram **NOT**
accurately drawn BC is parallel to AD

Find the size of the largest angle inside the trapezium.

○

(Total for Question 3 is 4 marks)

02 Interior and Exterior Angles of Polygons

For a regular polygon, go through the exterior angle: $360 \div n$. For any polygon, the interior angles add to $(n - 2) \times 180^\circ$.

WHEN YOU SEE → YOU DO

“each interior angle is 162° — how many sides?”

→ exterior 18° , $n = 360 \div 18 = 20$

“a 10-sided polygon ... find x ”

→ sum = $8 \times 180 = 1440^\circ$

“interior $(9x + 27)$, exterior $(2x - 23)$ ”

→ they add to 180

“Zak says the angles of a pentagon add to 450”

→ $(5 - 2) \times 180 = 540$

Exterior = $360 \div n$

Interior = $180 - \text{exterior}$

Sum = $(n - 2) \times 180$

WORKED MODEL • NOVEMBER 2025 P1H

A regular polygon has interior angle $(9x + 27)^\circ$ and exterior angle $(2x - 23)^\circ$. Find n .

1 $9x + 27 + 2x - 23 = 180 \Rightarrow x = 16$

interior + exterior = 180

2 exterior = $2(16) - 23 = 9^\circ$

substitute

ANSWER $n = 360 \div 9 = 40$

IDEA 02

Practise: Interior and Exterior Angles of Polygons

My questions first, then selected practice, then real past papers.

Q9 DR ESLAM Exterior first

CH 4 • TOPIC 2 3 marks

Each interior angle of a regular polygon is 156° . How many sides does it have?

Q10 DR ESLAM Sum of interior angles

CH 4 • TOPIC 2 2 marks

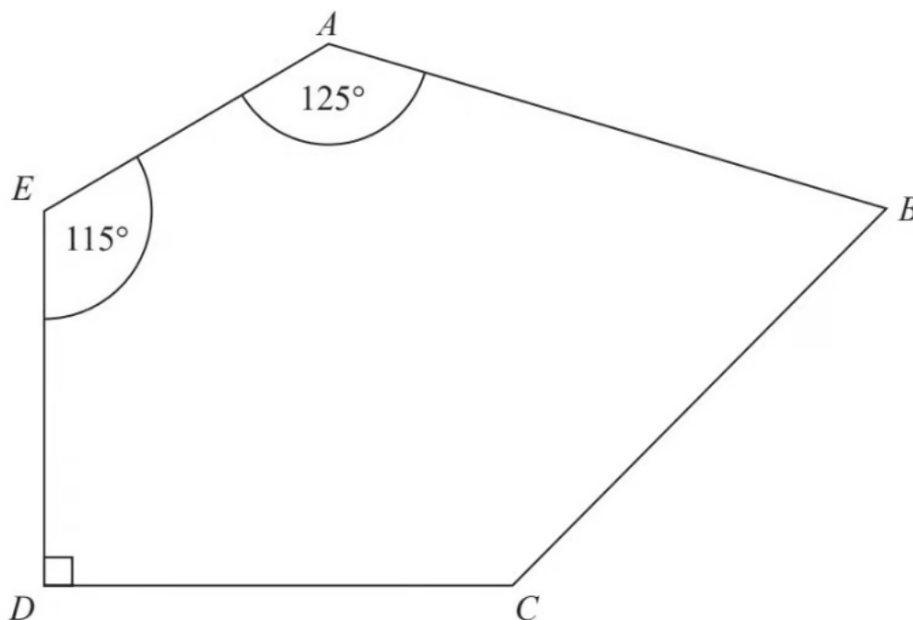
Work out the sum of the interior angles of a 9-sided polygon.

SELECTED PRACTICE

Q11 PRACTICE Practice · Medium

CH 4 · TOPIC 2 5 marks

9 $ABCDE$ is a pentagon.



Angle $BCD = 2 \times$ angle ABC .

Work out the size of angle BCD .

You must show all your working.

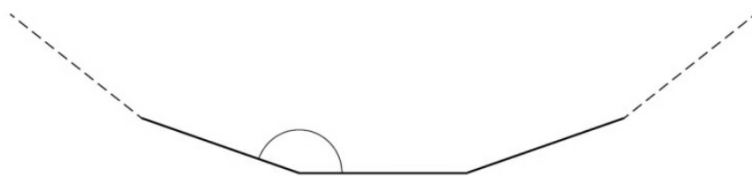
(5 marks)

Q12**PRACTICE**

Practice · Medium

CH 4 · TOPIC 2 **2 marks**

16 Part of a regular polygon with 15 sides is shown.



Not drawn
accurately

Work out the size of an interior angle.

.....degrees

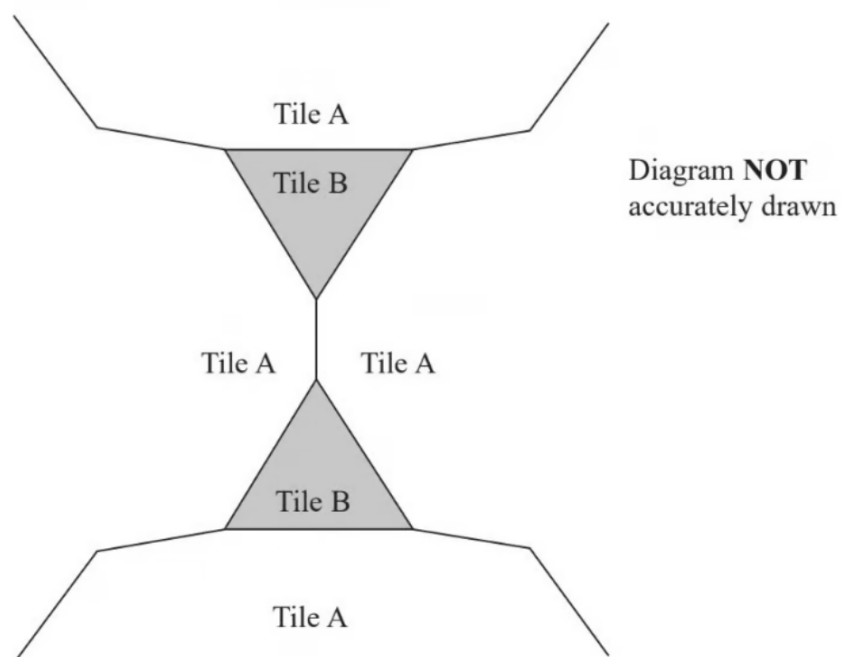
(2 marks)

Q13**PRACTICE**

Practice · Hard

CH 4 · TOPIC 2 4 marks

10 The diagram shows part of a pattern made from tiles.



The pattern is made from two types of tiles, tile A and tile B.

Both tile A and tile B are regular polygons.

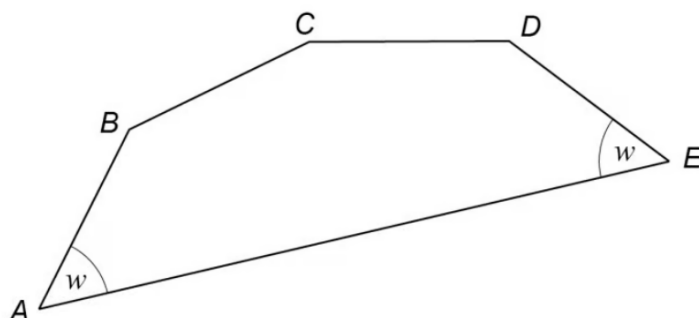
Work out the number of sides tile A has.

(4 marks)

Q14 PRACTICE Practice · Very Hard

CH 4 · TOPIC 2 **3 marks**

- 8 AB , BC , CD and DE are four of the sides of a regular decagon.



Not drawn
accurately

Work out the size of angle w .

.....degrees

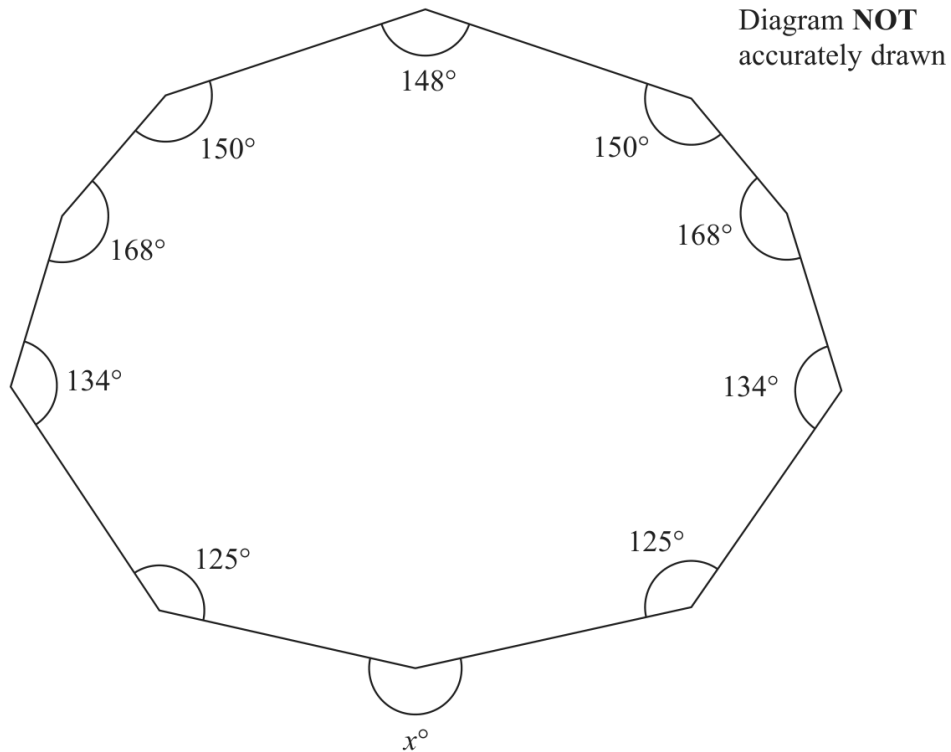
(3 marks)

PAST PAPERS

Q15 PAST PAPER Edexcel 4MA1 · Oct/Nov 2020 · Paper 1H · Q8

CH 4 · TOPIC 2 4 marks

8 Here is a 10-sided polygon.

Work out the value of x . $x = \dots\dots\dots$

(Total for Question 8 is 4 marks)

Q16 PAST PAPER Edexcel 4MA1 · June 2022 · Paper 1H · Q4

CH 4 · TOPIC 2 4 marks

- 4 The diagram shows parts of three regular polygons, **A**, **B** and **C**, meeting at a point.

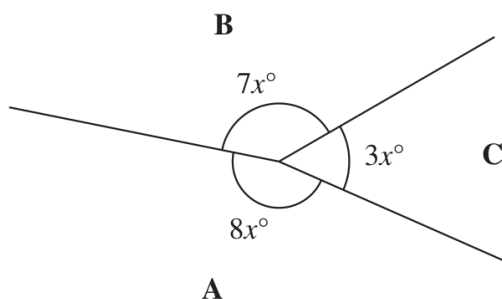


Diagram **NOT**
accurately drawn

Polygon **B** has n sides.

Work out the value of n .

$n = \dots\dots\dots$

(Total for Question 4 is 4 marks)

Q17**PAST PAPER**

Edexcel 4MA1 · June 2022 · Paper 1H · Q23

CH 4 · TOPIC 2 6 marks

23 A polygon has n sides, where $n > 5$

When arranged in order of size, starting with the largest number, the sizes of the interior angles of the polygon, in degrees, are the terms of an arithmetic sequence.

Here are the first five terms of this sequence.

177 175 173 171 169

Find the value of n

Show clear algebraic working.

Question 23 continued

$n = \dots\dots\dots$

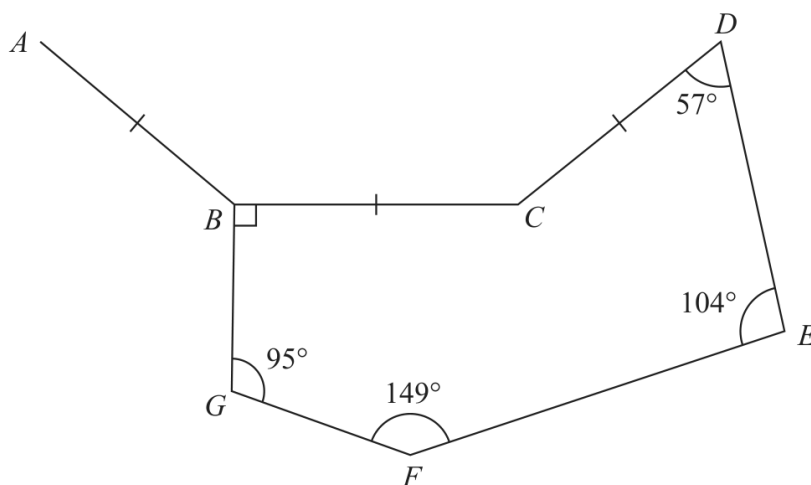
(Total for Question 23 is 6 marks)

Q18**PAST PAPER**

Edexcel 4MA1 · May 2023 · Paper 1HR · Q15

CH 4 · TOPIC 2 4 marks

15

Diagram **NOT**
accurately drawn $BCDEFG$ is a hexagon. AB , BC and CD are three sides of a regular n -sided polygon.Calculate the value of n

Show your working clearly.

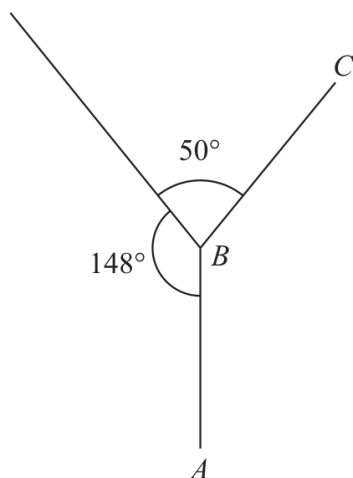
 $n = \dots\dots\dots$ **(Total for Question 15 is 4 marks)**

Q19**PAST PAPER**

Edexcel 4MA1 · June 2025 · Paper 1H · Q3

CH 4 · TOPIC 2 4 marks

3

Diagram **NOT**
accurately drawn

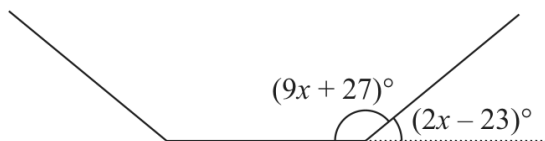
AB and BC are two sides of a regular polygon with n sides.

Work out the value of n
Show your working clearly.

 $n = \dots\dots\dots$ **(Total for Question 3 is 4 marks)**

Q20**PAST PAPER**

Edexcel 4MA1 · November 2025 · Paper 1H · Q12

CH 4 · TOPIC 2 4 marks**12**Diagram **NOT**
accurately drawnThe diagram shows part of a regular n -sided polygon withan interior angle of $(9x + 27)^\circ$ an exterior angle of $(2x - 23)^\circ$ Work out the value of n $n = \dots\dots\dots$ **(Total for Question 12 is 4 marks)**

03 Regular Polygons Joined Together

Find each polygon's interior angle first. Where they meet, the angles add to 360° . Equal sides from two polygons make isosceles triangles.

WHEN YOU SEE → YOU DO

“regular octagon and regular pentagon share a side”

→ 135° and 108°

angles around a point where polygons meet

→ 360° minus the interior angles

a triangle made by two polygon sides

→ isosceles: sides are equal

“O is the centre”

→ OD bisects the interior angle

sides	3	4	5	6	8	10
exterior	120	90	72	60	45	36
interior	60	90	108	120	135	144

WORKED MODEL • NOVEMBER 2024 P2H

Regular pentagon $ABCDE$ and regular hexagon $DEFGHI$ share DE . Find angle EAF .

1 $AEF = 360 - 108 - 120 = 132^\circ$

angles at a point

2 $EA = EF$ (equal sides)

isosceles triangle

ANSWER $AEF = \frac{180-132}{2} = 24^\circ$

IDEA 03

Practise: Regular Polygons Joined Together

My questions first, then selected practice, then real past papers.

Q21 DR ESLAM Two polygons

CH 4 • TOPIC 2 3 marks

A regular hexagon and a square meet at a point with a gap angle x . Find x .

SELECTED PRACTICE

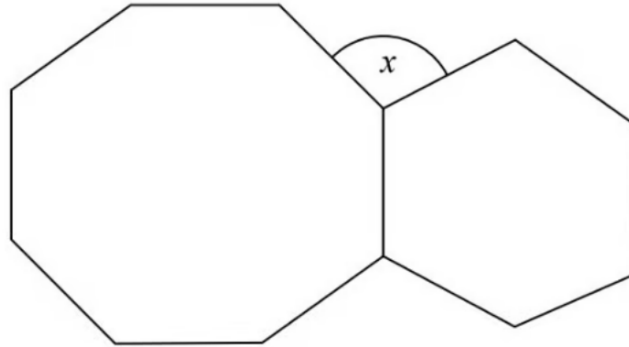
Q22

PRACTICE

Practice · Medium

CH 4 · TOPIC 2 3 marks

5



The diagram shows a regular octagon and a regular hexagon.

Find the size of the angle marked x .

You must show all your working.

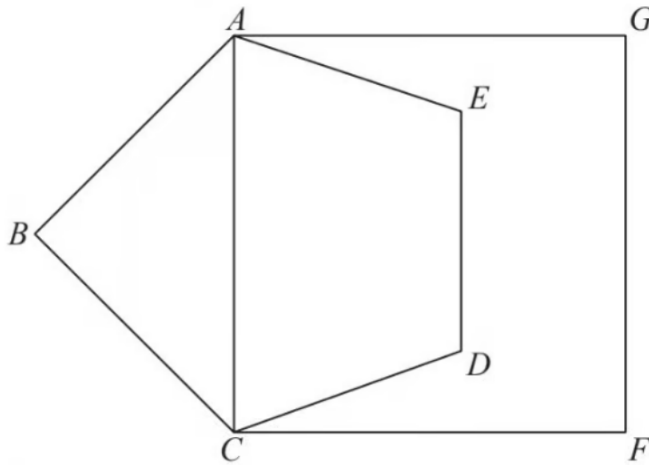
(3 marks)

Q23**PRACTICE**

Practice · Hard

CH 4 · TOPIC 2 4 marks

11

Diagram **NOT**
accurately drawn

$ABCDE$ is a regular pentagon.
 $ACFG$ is a square.

Work out the size of angle DCF .
You must show all your working.

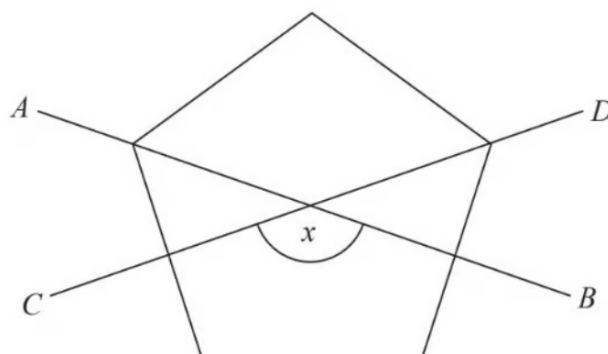
(4 marks)

Q24**PRACTICE**

Practice · Very Hard

CH 4 · TOPIC 2 4 marks

1

Diagram **NOT**
accurately drawn

The diagram shows a regular pentagon.

AB and CD are two of the lines of symmetry of the pentagon.

Work out the size of the angle marked x .

You must show all your working.

(4 marks)

Q25**PRACTICE**

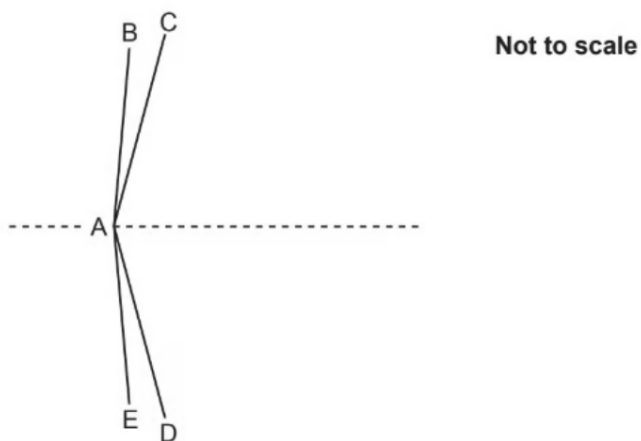
Practice · Very Hard

CH 4 · TOPIC 2 5 marks

12 Angle BAE is part of a regular 18-sided polygon.

Angle CAD is part of a regular 10-sided polygon.

The dashed line through A is a line of symmetry of both polygons.



Work out angle BAC.

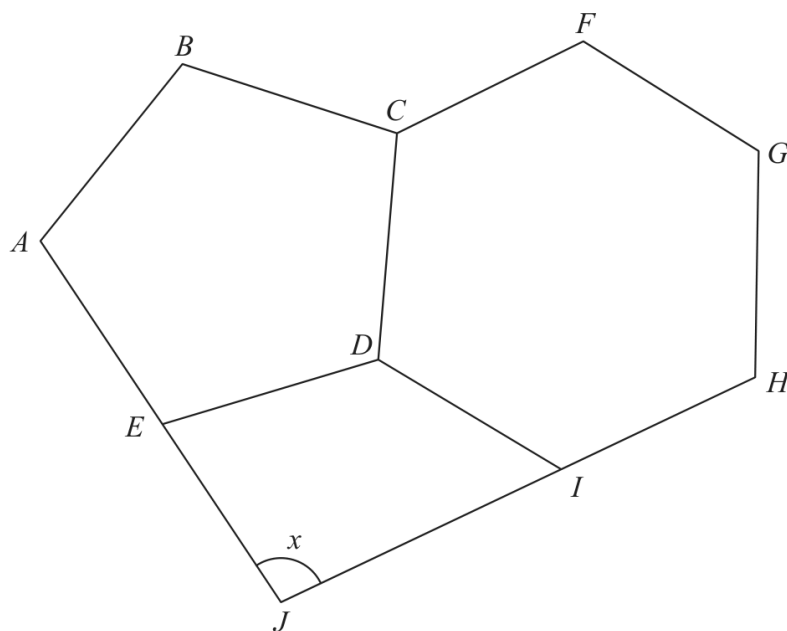
(5 marks)

PAST PAPERS

Q26 PAST PAPER Edexcel 4MA1 · January 2021 · Paper 1HR · Q8

CH 4 · TOPIC 2 5 marks

- 8 The diagram shows a regular pentagon, $ABCDE$, a regular hexagon, $CFGHID$, and a quadrilateral, $EDIJ$.

Diagram **NOT** accurately drawn

AEJ and HIJ are straight lines.

Work out the size of the angle marked x .
Show your working clearly.

○

(Total for Question 8 is 5 marks)

Q27

PAST PAPER

Edexcel 4MA1 · November 2021 · Paper 2H · Q5

CH 4 · TOPIC 2 5 marks

- 5 The diagram shows a regular octagon $ABCDHIJK$ and a pentagon $DEFGH$.

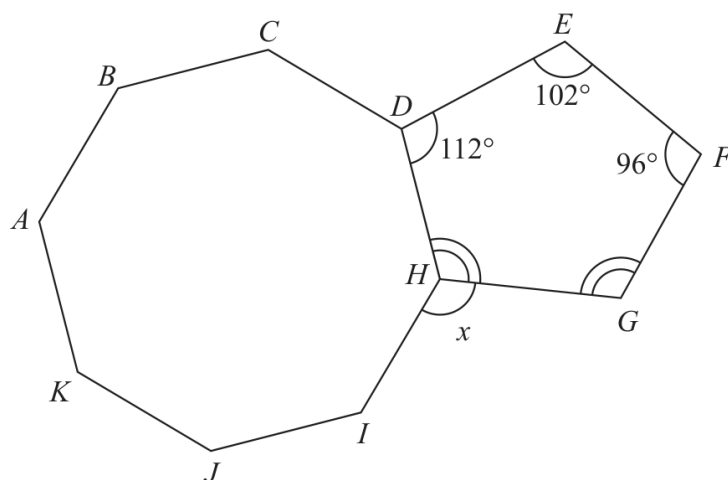


Diagram **NOT**
accurately drawn

Angle GHD = angle FGH .

Work out the size of the angle marked x .
Show your working clearly.

(Total for Question 5 is 5 marks)

Q28**PAST PAPER**

Edexcel 4MA1 · January 2022 · Paper 2HR · Q4

CH 4 · TOPIC 2 4 marks

- 4 The diagram shows a regular octagon $ABCDEFGH$ and a regular pentagon $ABIJK$

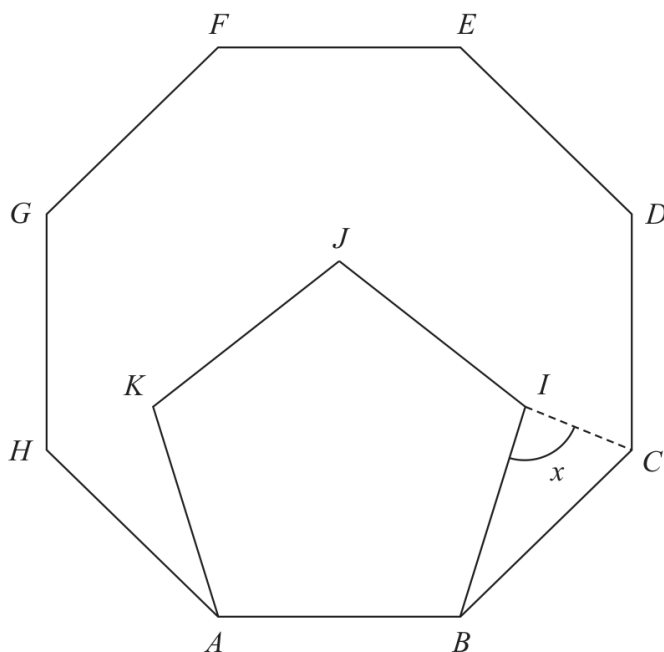


Diagram **NOT**
accurately drawn

Work out the size of the angle x

○

(Total for Question 4 is 4 marks)

Q29**PAST PAPER**

Edexcel 4MA1 · January 2022 · Paper 2H · Q11

CH 4 · TOPIC 2 4 marks

- 11 The diagram shows a regular 10-sided polygon, $ABCDEFGHIJ$

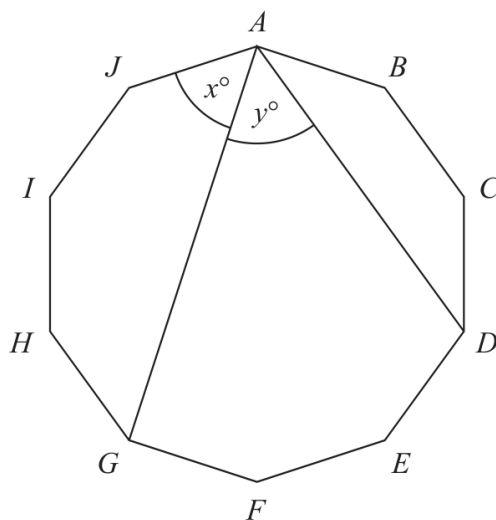


Diagram **NOT**
accurately drawn

Show that $x = y$

(Total for Question 11 is 4 marks)

Q30**PAST PAPER**

Edexcel 4MA1 · June 2023 · Paper 2H · Q11

CH 4 · TOPIC 2 3 marks

11 Here is a 9-sided regular polygon $ABCDEFGHIJ$, with centre O

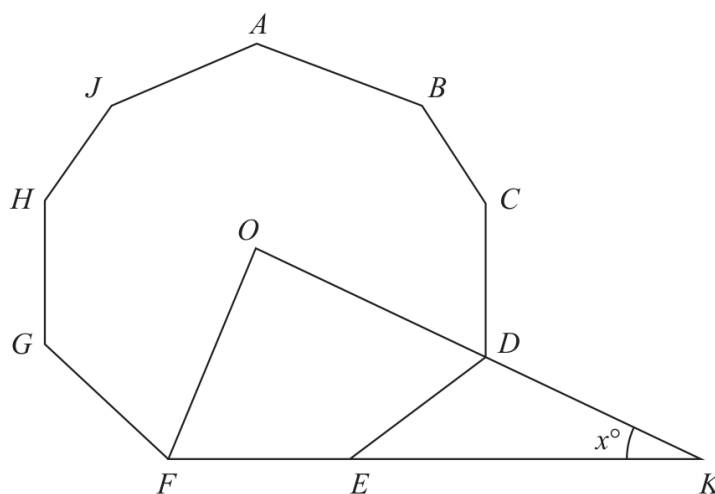


Diagram **NOT**
accurately drawn

ODK and FEK are straight lines.

Work out the value of x

$x = \dots\dots\dots$

(Total for Question 11 is 3 marks)

Q31 PAST PAPER Edexcel 4MA1 · May 2024 · Paper 1H · Q5

CH 4 · TOPIC 2 4 marks

- 5 In the diagram, $ABCDE$ is a regular pentagon.

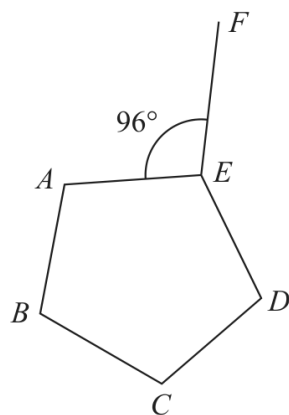


Diagram **NOT**
accurately drawn

Angle $AEF = 96^\circ$

Work out the size of the obtuse angle FED
Show your working clearly.

○

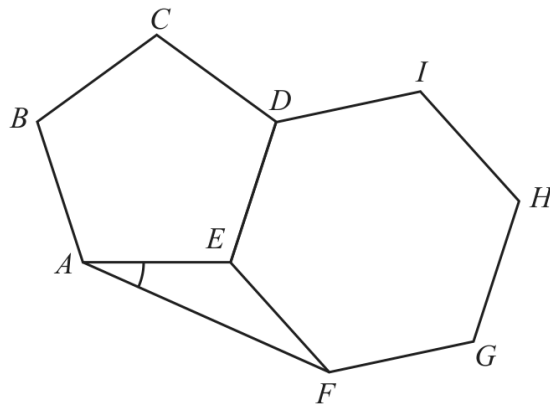
(Total for Question 5 is 4 marks)

Q32**PAST PAPER**

Edexcel 4MA1 · November 2024 · Paper 2H · Q11

CH 4 · TOPIC 2 5 marks

11

Diagram **NOT**
accurately drawn

$ABCDE$ is a regular pentagon.
 $DEFGHI$ is a regular hexagon.

AF is a straight line.

Work out the size of angle EAF

○

(Total for Question 11 is 5 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|------------------------|---|
| 01 | Angle facts | Alternate / corresponding equal, co-interior 180° ; write the reason. |
| 02 | Polygons | Exterior = $360 \div n$; interior = $180 - \text{exterior}$; sum $(n - 2)180$. |
| 03 | Joined polygons | Interior angles first; 360° at a point; isosceles triangles. |

BOTTOM LINE One angle at a time → a reason every time → polygons: exterior angle first.

Answers

Angles in Polygons and Parallel Lines • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • ANGLE FACTS AND PARALLEL LINES

- Q1** Dr Eslam • Parallel lines $180 - 64 = 116^\circ$ (co-interior angles add to 180°)
- Q2** Practice • Medium $x = 133^\circ$ (corresponding / angles on a line)
- Q3** Practice • Medium parallelogram: true; rectangle, rhombus, kite: may be true / not true (rhombus, kite: not true)
- Q4** Practice • Hard $BCD = 75^\circ$ so $ABC = 105^\circ$; $FBC = EFD = 35^\circ$ (alternate): $ABF = 105 - 35 = 70^\circ$
- Q5** Practice • Very Hard $x + 25 + 2x + 35 = 180 \Rightarrow x = 40$; $5x - 85 = 115 = 2x + 35$, corresponding angles equal $x = 24$
- Q6** Edexcel 4MA1 • January 2020 • Paper 2H • Q4
- Q7** Edexcel 4MA1 • May 2021 • Paper 2H • Q5 $EBC = 68^\circ$ (isosceles), $EGH = 68^\circ$ (corresponding): $JGH = 112^\circ$
- Q8** Edexcel 4MA1 • June 2023 • Paper 2H • Q3 $A + B = 180 \Rightarrow x = 23$: angles 65, 115, 79, 101: 115°

IDEA 02 • INTERIOR AND EXTERIOR ANGLES OF POLYGONS

- Q9** Dr Eslam · Exterior first
 exterior 24° : $360 \div 24 = 15$
- Q10** Dr Eslam · Sum of interior angles
 $(9 - 2) \times 180 = 1260^\circ$
- Q11** Practice · Medium
 $ABC + BCD = 210$, $3ABC = 210$: $BCD = 140^\circ$
- Q12** Practice · Medium
 156°
- Q13** Practice · Hard
 tile B equilateral (60°): $2A + 60 = 360$, $A = 150^\circ$
 $\rightarrow 12$ sides
- Q14** Practice · Very Hard
 $3(144) + 2w = 540$: $w = 54^\circ$
- Q15** Edexcel 4MA1 · Oct/Nov 2020 · Paper 1H · Q8
 $x = 138$
- Q16** Edexcel 4MA1 · June 2022 · Paper 1H · Q4
 $18x = 360$, $x = 20$: B interior $140^\circ \rightarrow n = 9$
- Q17** Edexcel 4MA1 · June 2022 · Paper 1H · Q23
 $\frac{n}{2}(354 - 2(n - 1)) = 180(n - 2)$: $n = 18$
- Q18** Edexcel 4MA1 · May 2023 · Paper 1HR · Q15
 $(270 - l) + (360 - l) = 315$, $l = 157.5^\circ$: $n = 16$
- Q19** Edexcel 4MA1 · June 2025 · Paper 1H · Q3
 interior 162° , exterior 18° : $n = 20$
- Q20** Edexcel 4MA1 · November 2025 · Paper 1H · Q12
 $11x + 4 = 180$, $x = 16$: exterior $9^\circ \rightarrow n = 40$

IDEA 03 · REGULAR POLYGONS JOINED TOGETHER

- Q21** Dr Eslam · Two polygons $360 - 120 - 90 = 150^\circ$
- Q22** Practice · Medium $360 - 135 - 120 = 105^\circ$
- Q23** Practice · Hard $ACD = 108 - 36 = 72^\circ$; $DCF = 90 - 72 = 18^\circ$
- Q24** Practice · Very Hard $x = 144^\circ$
- Q25** Practice · Very Hard $\frac{160}{2} - \frac{144}{2} = 8^\circ$
- Q26** Edexcel 4MA1 · January 2021 · Paper 1HR · Q8 $x = 360 - 72 - 60 - 132 = 96^\circ$
- Q27** Edexcel 4MA1 · November 2021 · Paper 2H · Q5 pentagon: $2y = 230$, $y = 115$: $x = 360 - 135 - 115 = 110^\circ$
- Q28** Edexcel 4MA1 · January 2022 · Paper 2HR · Q4 $IBC = 135 - 108 = 27^\circ$, isosceles: $x = 76.5^\circ$
- Q29** Edexcel 4MA1 · January 2022 · Paper 2H · Q11 triangles AJG and ADG fill the angle: each cuts 3 equal edges, so $x = y = 54^\circ$
- Q30** Edexcel 4MA1 · June 2023 · Paper 2H · Q11 $ODE = 70^\circ$, $DEK = 40^\circ$: $x = 30$
- Q31** Edexcel 4MA1 · May 2024 · Paper 1H · Q5 $360 - 108 - 96 = 156^\circ$
- Q32** Edexcel 4MA1 · November 2024 · Paper 2H · Q11 $AEF = 132^\circ$, isosceles: $EAF = 24^\circ$